

Formula Sheet

Time Value of Money:

$$\text{Present value, } PV = \frac{FV}{(1+r)^t}$$

$$\text{Future value, } PV * (1 + r)^t$$

$$T = \frac{\log\left(\frac{FV_T}{PV}\right)}{\log(1 + r)}$$

$$r = \left(\frac{FV_T}{PV}\right)^{\frac{1}{T}} - 1$$

$$\text{Compounding: } FV_T = C_0 * \left(1 + \left(\frac{r}{m}\right)\right)^{m \times T}$$

$$\text{Continuous Compounding: } FV_T = C_0 * e^{rT}$$

$$\text{EAR} = \left(1 + \frac{r}{m}\right)^m - 1, m = \# \text{ of compounding periods}$$

PV = Present Value

FV = Future Value

T = Time Periods

r = discount rate

C₀ = Initial Investment at time 0

m = number of compounding periods

Annuities and Perpetuities:

$$\text{PV of Annuity, } PV = \frac{C}{r} \left[1 - \frac{1}{(1+r)^t}\right], \text{ Growing: } PV = \frac{C}{r-g} \left[1 - \left(\frac{1+g}{1+r}\right)^T\right]$$

$$\text{FV of annuity, } FV = C \left[\frac{(1+r)^t - 1}{r}\right]$$

$$\text{PV of Annuity due: } PV_{\text{annuity due}} = PV_{\text{annuity}} * (1 + r)$$

$$\text{Perpetuity: } PV = \frac{C}{r}, \text{ Growing: } PV = \frac{C_{T+1}}{r-g}$$

PV = Present Value

FV = Future Value

T = Time Periods

r = discount rate

g = growth rate

C = Cash Flow

Bond Value:

$$PV = C \left[\frac{1}{r} - \frac{1}{r(1+r)^t} \right] + \frac{F}{(1+r)^t}$$

$$\text{Holding Period Return: } \frac{P_n - P_o + \sum \text{Income}_n}{P_o}$$

$$\text{Forward Rate Formula: } f_n = \frac{(1 + r_n)^n}{(1 + r_{n-1})^{n-1}} - 1$$

PV = Present Value or Market Price

FV = Face Value

T = Time Periods

r = discount rate

C = Cash Flow

P_n = New Price

P_o = Old Price

f_n = forward rate

Stock Value:

$$P_o = \frac{D_{t+1}}{r-g}, P_o = \frac{D_1}{r}, P = \frac{C}{r-g} \left[1 - \left(\frac{1+g_1}{1+r} \right)^N \right] + \frac{\frac{Div_{N+1}}{r-g_2}}{(1+r)^N}$$

$$g = RR \times ROE$$

$$r = \frac{Div_{N+1}}{P_t} + g$$

$$P_{per\ share} = \frac{EPS}{r} + NPVGO(per\ share) = \frac{EPS}{r} + \frac{NPV}{\#of\ Shares}$$

$$V_o = \frac{FCF_1}{1 + r_{WACC}} + \frac{FCF_2}{(1 + r_{WACC})^2} + \dots + \frac{FCF_N}{1 + r_{WACC}} + \frac{V_N}{(1 + r_{WACC})^N}$$

$$V_N = \frac{FCF_{N+1}}{r_{WACC} - g} = \frac{FCF_N(1 + g)}{r_{WACC} - g}$$

$$\text{Equity Value Per Share} = \frac{V_o + Cash_0 - Debt_0}{Shares\ Outstanding_0}$$

$$\text{Enterprise Value} = \text{Equity Value} + \text{Debt} - \text{Cash}$$

P_o = Present Value or Market Price

V_o = Discounted Value of FCF

N = Time Periods

r = discount rate

g₁ = first growth rate

g₂ = second growth rate

C = First Dividend at the initial growth rate
 Div_{n+1} = First Dividend at the new growth rate
 FCF = Free Cash Flow
 RR = Retention Ratio
 ROE = Return on Equity
 EPS = Earnings Per Share

Project Evaluation Formulas:

NPV = Pv of all cash flows

$NPV = PV((Rev - Costs)(1 - T)) + PV(ECF) + PV(Tax Shield) - Cost of Asset$

$PV(TS) = \left(T_c * \left(\frac{C_o}{N} \right) \right) * A_r^N$, where A_r^N is the annuity factor

$PV(CCATS) = \left(\frac{T_c * d * C_o}{k + d} * \frac{1 + 1.5k}{1 + k} \right) - \left(\frac{T_c * d * S}{k + d} * \frac{1}{(1 + k)^N} \right)$

$NPV = \sum_{t=0}^N \frac{CF_t}{(1 + r)^t} - C_0$

$IRR = \sum_{t=0}^N \frac{CF_t}{(1 + IRR)^t} - CF_0 = 0$

$PI = \sum_{t=1}^T \frac{\frac{CF_t}{(1 + r)^t}}{C_0}$

$EAB = NPV / A_r^T$

NPV = Net Present Value
 IRR = Internal Rate of Return
 PI = Profitability Ratio
 EAB = Equivalent Annual Benefit
 $k = r$ = discount rate
 d = CCA rate
 S = Salvage Value

Average Return of Stock:

$E(R_A) = \frac{R_1 + \dots + R_T}{T}$

R or r = Return

T = Number of Returns

Geometric Return of Stock:

$$E(R_A) = \prod_{t=1}^T (1 + r_t)^{\frac{1}{T}} - 1$$

Expected Return of Stock:

$$E(R_A) = r_1 p_1 + r_2 p_2 + r_3 p_3 + \dots + r_n p_n$$

P or p = probability

Standard deviation of Stock:

Expected:

$$\sigma_A = \sqrt{P_1(r_1 - E(R_A))^2 + P_2(r_2 - E(R_A))^2 + P_3(r_3 - E(R_A))^2}$$

Historical:

$$\sigma_A = \frac{\sqrt{((r_1 - E(R_A))^2 + (r_2 - E(R_A))^2 + (r_3 - E(R_A))^2)}}{T - 1}$$

Expected Return of Portfolio (two assets):

$$E(R_p) = \omega_A E(R_A) + \omega_B E(R_B)$$

$$\omega_i = \frac{H_i}{\sum_{k=1}^N H_k}$$

H_i = amount of asset held

Standard deviation of Portfolio (two assets):

$$\sigma_p = \sqrt{\omega_A^2 \sigma_A^2 + \omega_B^2 \sigma_B^2 + 2\omega_A \omega_B \text{cov}_{A,B}}$$

$$\sigma_p = \sqrt{\omega_A^2 \sigma_A^2 + \omega_B^2 \sigma_B^2 + 2\omega_A \omega_B \sigma_A \sigma_B \rho_{A,B}}$$

Covariance, correlation, and beta (two stocks):

$$\text{cov}_{ab} = \sigma_{a,b} = \sum_{i=1}^N (r_{ai} - E(r_a))(r_{bi} - E(r_b))P_i$$

$$\text{corr}_{a,b} = \rho_{a,b} = \frac{\sigma_{a,b}}{\sigma_a * \sigma_b}$$

$$\beta_j = \frac{Cov_{j,M}}{\sigma_M^2}$$

Portfolio Beta:

$$\beta_p = \omega_A \beta_A + \omega_B \beta_B + \dots + \omega_N \beta_N$$

CAPM :

$$E(R_j) = R_F + \beta_j(E(R_m) - R_F)$$

$R_f = \text{risk free rate}$

$R_m = \text{market return}$

Weighted-Average Cost of Capital:

$$WACC = \omega_D r_D (1 - T) + \omega_{ps} r_{ps} + \omega_s r_s, \quad WACC = \omega_D r_D (1 - T) + \omega_E r_E$$

$r_E \text{ and } r_D = \text{cost of equity and debt}$

$$\text{Sharpe Ratio} = \frac{E(r_i - r_f)}{\sigma_i}, \text{CML Equation: } E(r_p) = r_f + \frac{(E(r_k) - r_f)}{\sigma_k} \sigma_p$$

Equity and Asset Betas:

$$\beta_A = \left(\frac{E}{D + E} \right) \beta_E$$

$$\beta_E = \left(1 + \frac{D}{E} \right) \beta_A$$